W-G-663B
October 30, 1970
SUPERSEDING
Fed. Spec. W-G-663A
July 30, 1954

FEDERAL SPECIFICATION

GRINDER, ELECTRIC, PORTABLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable electric grinders.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Sizes. Portable electric grinders shall be of the following sizes, as specified (see 6.2):

Size 4 - 4-inch diameter grinding wheel.
Size 5 - 5-inch diameter grinding wheel.
Size 6 - 6-inch diameter grinding wheel.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 5130

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Military Specifications:

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1337 - Electric Portable Hand Tools General Suppression System Design Requirements.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids, or request for proposal shall apply:

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies shall be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

American National Standards Institute (ANSI), Inc. Standards:

B3.5 - Tolerances for Ball and Roller Bearings.
B3.6 - Boundary Dimensions for Ball and Roller Bearings.
B3.8 - Mounting Ball and Roller Bearings.
B5.38 - Driving and Spindle Ends for Portable Air and Electric Tools.
B7.1 - Use, Care, and Protection of Abrasive Wheels, The Safety Code for.
B74.2 - Standard Shapes and Sizes of Grinding Wheels.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

Underwriters' Laboratories (UL), Inc. Standards For Safety:

No. 45 - Electric Tools.
No. 498 - Attachment Plugs and Receptacles.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, Long Island, N.Y. 11746; or 1655 Scott Boulevard, Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of grinders otherwise conforming to this specification.

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3.2 First article inspection. When specified (see 6.2), the contractor shall furnish a completely fabricated portable electric grinder for first article inspection in accordance with section 4. The sample shall be a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used in production of items to fulfill the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirement of this specification.

3.3 Material. All material used in the fabrication of portable electric grinders shall be new and of a quality necessary to produce a machine that meets the requirements described herein. Magnesium materials shall not be used.

3.3.1 Construction. The grinder shall be free of any characteristics or defects that will prevent the grinder from passing any of the examinations and test specified in section 4. The grinder shall be complete so that when connected to the specified source of power, it can be used for its designed operation.

3.3.2 Castings. All castings shall be free from defects and mismatching. Reclamation of any parts of the grinder by such processes as welding, peening, plugging, or filling with solders or metallic pastes will not be permitted.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or permanently deformed.

3.3.4 Threads. All threaded parts shall be in the inch system and shall conform to H28.

3.3.5 Surface finishes. Exposed surfaces shall be smooth, and edges shall be rounded or beveled. Surfaces shall be finished according to the manufacturer's standard practice.

3.4 Design. The design shall be such as to provide convenient and safe operation. All parts subject to wear, breakage, and distortion shall be easily accessible for adjustment, replacement or repair.

3.4.1 Safety devices. The grinders shall be fitted with safety devices or covering for all parts that present a safety hazard. These devices shall include a guard for the abrasive wheel. All guards shall be designed to provide access to the guarded parts. Ventilating openings shall conform to the requirements of UL45. Tools with suppression systems shall bear a permanent label warning of the necessity for using the grounding connection to avoid shock.

3.4.2 Wheel caution. In addition to the nameplate data specified herein, the grinder shall be permanently marked with the following, or equivalent:

CAUTION: Use only wheels marked at or over actual free running R.P.M. of the tool.

3.4.3 Sizes and capacities. Grinders furnished shall be similar to figure 1 and comply with capacities and dimensions as listed in table I for the size of grinders specified (see 6.2).

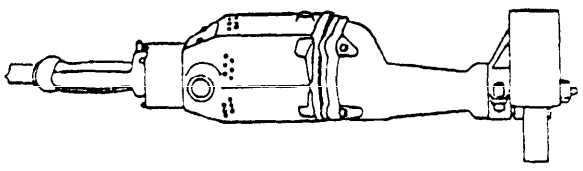


FIGURE 1. Grinder, electric, portable.

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TABLE I. Capacities and dimensions

Requirements	Size 4	Size 5	Size 6
Wheel capacity			
Diameter, inch, (max.)	4	5	6
Wheel face, width, inch	3/4	3/4	1
Arbor hole, inch, nominal (minimum)	1/2	1/2	5/8
Spindle-diameter inch (minimum)	1/2	1/2	5/8
Thread, inch	1/2-13UNC	1/2-13UNC	5/8-11UNC
Wheel speed			
No-load, R.P.M. (max.)	6200	4900	4100
1/ Full-load, R.P.M. (min.) at rated horsepower	3500	2700	2300
Horsepower, rated (min.) delivered at wheel end of spindle	0.33	0.43	0.75
Length - Overall, inch (max.)	22	24	26
Weight - Pounds net, (max.) including wheel guard and cable	16	20	26

1/ Based upon 6500 surface feet per minute inorganic bond-high strength abrasive wheel. (B7.1).

3.4.4 Lubrication. All bearings (except sealed for life type), mating gears, and sliding parts shall be provided with means for lubrication. Pressure lubrication fittings shall not be used on motor bearings. Gears shall operate in a lubricant which shall not enter the motor frame or leak from the gear casing.

3.4.5 Interchangeability. All parts shall be manufactured to such standards that will permit replacement or adjustment without modification of parts or grinder.

3.4.6 Conformance to standards of Underwriters' Laboratories, Inc. The contractor shall submit to the contracting agency proof that the basic grinder he proposes to supply under this specification conforms to the requirements of the UL45. The label, or listing with reexamination of the UL may be accepted as evidence that the grinder conforms to this requirement. In lieu of the label, the contractor may submit independent proof, satisfactory to the contracting agency, that the grinder conforms to the applicable requirements of the published standards, including methods of tests of the UL noted above. Compliance with requirements of UL regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his grinder. In event of difference between any of the requirements of this specification and those of UL, the requirements of this specification shall govern.

3.5 Components. The grinders furnished under this specification shall consist of housings, handle, motor, switch, wiring, cable and plug, gears and shafts, spindle, bearings, wheel guard, and grinding wheel.

3.5.1 Housings. Housings shall be of a material of suitable characteristics to withstand stresses imposed by normal usage.

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3.5.1.1 Handles. Grinders shall have a straight type handle at the end opposite the grinding wheel. Handle shall house the switch and be designed to provide a convenient grip for the operator. The spindle housing shall provide a comfortable gripping surface. Gripping surfaces shall be convenient for two-handed operation.

3.5.2 Motors. Motors shall be universal series wound or compensated series wound designed to operate on either 115 volt direct current or 115 volt, single phase, 25 to 60 cycle alternating current. Motors shall conform to UL45 requirements for electric tools.

3.5.2.1 Rotors and fans. Rotors and fans shall be balanced dynamically, within the standard limits specified commercially for the motor size used.

3.5.2.2 Commutators. Commutators shall be made of hard drawn, hard rolled or drop forged copper segments. Commutators shall be of "V" ring and sheet mica or molded construction. Commutators shall withstand maximum no-load r.p.m. of motor radial displacement of commutator segments.

3.5.3 Switch. Grinders shall be provided with a conveniently located switch of the quick break design. Switches shall release automatically and shall have a lock for holding in the "on" position. Switches shall have a current rating of not less than the rated load rating of the grinder. Unless otherwise specified (see 6.2), switches shall be rated AC/DC. Switches shall comply with the requirements of UL45.

3.5.4 Wiring. All wiring, flexible cord, attachment plug cap and grounding means shall be in accordance with UL45. The cord shall have an attachment plug cap and unless otherwise specified (see 6.2), a minimum free length of 9 feet measured from the point of entry into the grinder. Attachment plug cap shall conform to UL498 requirements for 2-pole, 3-wire grounding devices rated 15 ampers at 125 volts.

3.5.5 Gears and shafts. Gears and shafts shall be fabricated from suitably hardened alloy steel. Gears shall have machine cut or formed teeth and shall be adequately supported. Spindle shaft shall be supported by at least two bearings to prevent deflection. Gears and shafts shall be fully enclosed.

3.5.6 Spindle. Spindle shall be fabricated from suitably hardened alloy steel and shall be machined to the diameter specified in table I for the size grinder specified. Spindle shall be mounted on ball or roller bearing with integral or separate shielding and sealed to prevent leakage of lubricant or entrance of contaminants. Spindle end threads and dimensions shall conform to B5.38 for straight wheel grinders.

3.5.7 Bearings. Bearings shall be either ball or roller and shall conform to dimensions and tolerances of ANSI Standards for the type and size bearing used. Bearings shall be of the dustproof sealed type or be mounted in a dustproof housing. Steel inserts may be omitted in the mounting when sufficient contact area is provided on the mounting surface. Roller bearings which are designed for radial loads shall not be used to absorb thrust loads. Ball and roller bearings shall conform to B3.5 for tolerances, B3.6 for boundary dimensions, and B3.8 for mounting requirements.

3.5.8 Wheel guard. Each grinder shall be fitted with a wheel guard which will protect the operator from grinding debris or broken pieces of abrasive wheel. Safety guards shall conform to B7.1 for wheel guards 8 inches in diameter and smaller.

3.5.9 Grinding wheel. Each grinder shall be furnished with a grinding wheel of the dimensions specified in table I for the size specified. Wheel shall be square faced, 24 to 36 grit, vitrified or resinoid bond, and shall conform to type I shape of B74.2. The grinding wheel shall be mounted on the grinder clamped between two flange washers of identical outside diameter. Compressible washers (blotters) shall be used between the wheel and each flange. Flange washers and blotters shall conform to B7.1 for the type and size wheels provided. Each grinder shall be provided with a wrench for removing the wheel.

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3.6 Identification plate. Each grinder shall be provided with a corrosion-resistant nameplate securely fastened to or as an integral part of the housing. Nameplate shall be permanently and legibly marked to include the following:

(a) Manufacturer's name or trademark.
(b) Manufacturer's type, model, or serial number.
(c) Voltage AC/DC
(d) Full-load current-amperes.
(e) No-load r.p.m.

3.7 Handbook of instruction. Unless otherwise specified (see 6.2), the manufacturer shall furnish one copy of his regular instruction book and parts list with each grinder. The instruction book shall include all illustrations, instructions and other necessary data for safe operation, maintenance, adjustment, and repair of the tool. A parts list and ordering instructions shall be included.

3.8 Lifting eye. When specified (see 6.2), each grinder shall be provided with a lifting eye for aerial suspension. Eye shall be positioned for balance.

3.9 Accessories. In addition to the accessories specified herein, each grinder shall be furnished with such maintenance tools, accessories, and extra parts as normally furnished commercially.

3.10 Stand. When specified (see 6.2), a stand shall be provided for converting the portable grinder into a fixed bench grinder. Stand shall be of such construction that the grinder can be securely held in a rigid position. Stand shall include an adjustable tool rest and means designed for bench mounting.

3.11 Electromagnetic interference suppression. When specified (see 6.2), each grinder shall be electromagnetic interference suppressed in accordance with the requirements of MIL-STD-1337.

3.12 Fungus proofing. When specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, with the following exceptions:

(a) Components or circuit elements which are inherently fungus and water resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied to give a minimum dry film thickness of 0.001 inch to component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.13 Workmanship. Workmanship of portable electric grinders shall be of the quality prevailing among manufacturers normally producing electric grinders of the size specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection.
(b) Quality conformance inspection.

4.2.1 First article inspection. First article inspection shall be applied to the first article submitted in accordance with 3.2. Failure of the first article to pass the examination on one or more preproduction tests shall be cause for rejection. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.4.1 and all tests in 4.5.

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4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All grinders of the same size offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sample. A random sample shall be selected from each lot in conformance with MIL-STD-105 at inspection level II.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each sample selected in accordance with 4.3.2. Quality conformance inspection shall consist of the following:

(a) Examination (4.4.1).
(b) Tests (4.4.2).
(c) Inspection of preservation, packaging, packing, and storage (4.6).

4.4.1 Examination. Each sample selected in accordance with 4.3.2 shall be examined for design construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table II. The acceptable quality level (AQL) for major defects shall be 2.5 percent defective and for minor defects 4.0 percent defective. Any deviation from the requirements specified shall be cause for rejection. Rejected samples shall not be included in lot for delivery and if the number of rejects exceeds the acceptance number for that sample, the lot represented shall be rejected.

TABLE II. Classification of defects

Examine	Defects	AQL percent defective		
		Critical 0.10	Major 2.5	Minor 4.0
Design	Evidence of electric shock or fire hazard.	X		
	Capacity or size of grinder less than specified.		X	
Material	Not in accord, when specified.		X	
Construction	Nameplate and safety plate nonconforming.		X	
	Handle and spindle hand hold not comfortable or suitable for 2-hand control.		X	
	Switch not type specified.		X	
	Locking device for switch not included or not operating automatically.		X	
	Cable and plug not as specified.		X	
	Wheel guard missing or not of the style specified.		X	
	Accessories not as specified in order.		X	
	Workmanship not of quality expected of good commercial manufacturing procedures.		X	
	Spindle size and threads not as specified.		X	
Technical Manuals	Instruction books and parts lists missing.			X
	Instruction book data not as specified.			X

4.4.2 Tests for quality conformance. Each sample selected in accordance with 4.2.2 shall be tested for compliance with the requirements of this specification. Nonconforming grinders shall not be offered for delivery and if the number of defective units exceeds the acceptable number for the samples tested, the lot represented shall be rejected. Quality conformance tes consist of the tests specified in 4.5.1 through 4.5.3.

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4.5 Tests.

4.5.1 Electromagnetic interference suppression test report. When electromagnetic interference suppression is required, the manufacturer shall furnish: (1) a certified test report showing the grinders supplied have been tested and are within the limits specified for portable electric tools in accordance with MIL-STD-1337 or (2) certification that the electromagnetic suppression network in the grinders offered is identical to that of grinders previously tested and accepted. The electromagnetic interference suppression system shall be disconnected during the tests of 4.5.2.

4.5.2 Motor and electrical equipment tests. Each sample grinder shall be tested to determine compliance with the requirements of UL45 for industrial tools, for the following: (test voltages are RMS values unless otherwise stated)

(a) Continuity of grounding connections.
(b) Insulation resistance.
(c) Dielectric strength.

4.5.3 No-load test. Each grinder shall be operated at no-load r.p.m. at rated voltage and ambient temperature. When operating temperatures are constant each grinder shall be operated for a period of one hour. No-load spindle r.p.m. determinations shall be made at the start and termination of the one-hour period. No-load r.p.m. shall not vary more than ± 10 percent. During the test the grinder shall be checked for switch operation, excessive noise, vibration, loss of lubricant and indication of overheating.

4.5.4 Rated-load test. For pre-production test only, after no-load operation the grinder shall be operated at rated-load. When operating temperatures are stable the following determinations shall be made:

(a) Current Input (Rated-load current, amperes at 60 cycles A.C. and 120 volts).
(b) Spindle speed (r.p.m.).
(c) Power at spindle (HP).
(d) Temperature rise.

After rated-load test, operational components including motor parts, bearings, and gears shall be examined for wear, insecure fastenings, displacement of commutator segments, and evidence of lubricant leakage.

4.6 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40 for grinders, electric, portable.

6. NOTES

6.1 Intended use. The heavy duty production type grinders as furnished under this specification are hand guided, hand pressure applied, portable electric aerial grinders for use in foundries, forge shops, welding shops, and general grinding operations in the removal of burrs, fins, flash, excess weld material, sprues, and similar hand grinding operations. When mounted on a stand, they can be used as a bench grinder.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of grinder required (see 1.2.1 and 3.4.3).
(c) Cord length, if different (see 3.5.4).
(d) Handbook requirements when different (see 3.7).
(e) Lifting eye, if required (see 3.8).
(f) Stand, if required (see 3.10).
(g) Electromagnetic interference suppression when required (see 3.11).
(h) Fungus proofing, if required (see 3.12).
(i) First article tests, if different (see 4.2.1).
(j) Switch rating if other than specified (see 3.5.3).

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6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 First article approval. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article approval as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

MILITARY INTEREST:

Custodians:

Army - GL
Navy - SH
Air Force - 84

Review activities:

Army - EL, GL
Navy - SH
Air Force - 84

User activities:

Army - CE
Navy - AS, MC

Preparing activity:

Navy - SH

CIVIL AGENCY INTEREST:

GSA-FSS

Project No. 5130-0071

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rain specified for all input electrical output requirements be